

MAT1002: Calculus II L01

Quiz 4 — 2025-2026, Term 2

Tuesday, 21 Apr 2026

1. (a) (4 points) Evaluate the double integral $\iint_D (x + 2y) dA$, where D is the region bounded by the parabolas $y = 2x^2$ and $y = 1 + x^2$.
(b) (6 points) Evaluate $\iiint_E z dV$, where E is the solid tetrahedron bounded by the planes $x = 0, y = 0, z = 0$ and $x + y + z = 1$.
2. (a) (3 points) A solid E lies inside the cylinder $x^2 + y^2 = 1$, below the plane $z = 4$, and above the paraboloid $z = 1 - x^2 - y^2$. The density at a point $P(x, y)$ is $K\sqrt{x^2 + y^2}$ for some constant K . Find $\text{mass}(E)$.
(b) (7 points) Find the volume of the solid that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = z$.
3. (10 points) Evaluate the integral

$$\iint_D e^{\frac{x+y}{x-y}} dA,$$

where D is the trapezoidal region with vertices $(1, 0), (2, 0), (0, 1)$ and $(0, 2)$.